



School of Computer Sciences

CPC351/CPM351 Principles of Data Analytics

Academic Session: Semester 1, 2024/2025

Assignment 02 – Data Exploration and Visualization

I. Dataset

Download the following datasets from eLearn@USM:

- Vehicle Registration Dataset: cars_2019.csv, cars_2020.csv, cars_2021.csv, cars_2022.csv, cars_2023.csv, and cars_2024.csv.
- Daily Public Transport Ridership Dataset: ridership_headline.csv.

The datasets were taken from Malaysia's Official Open Data Portal (data.gov.my). You need to use the dataset downloaded from eLearn@USM, not the one downloaded from the data.gov.my, as the dataset on the open data portal will keep changing.

The datasets are made open under the Creative Commons Attribution 4.0 International License (CC BY 4.0). Please refer to <https://creativecommons.org/licenses/by/4.0/> for more information on the license.

II. Vehicle Registration Dataset

The vehicles dataset tabulates actual transactional records of all vehicles registered with the Road Transport Department (JPJ), from 01 Jan 2019 to 30 Nov 2024). Kindly refer to the URL https://data.gov.my/data-catalogue/registration_transactions_car to get further information on the dataset. Load the data into R environment. Then, perform data pre-processing and data cleaning and answer the following questions.

1. Using appropriate visuals, show the number of vehicles registered at each state JPJ office (including the Rakan Niaga) for the years of 2019, 2020, 2021, 2022, 2023, and 2024.
2. Create a visual to identify the top five brands of registered vehicles in Pulau Pinang for the years 2019, 2020, and 2021.
3. Based on the given datasets (excluding the Rakan Niaga), what is the favourite vehicles colour for the following regions:
 - a. Northern region of Peninsular Malaysia: Perlis, Kedah, Pulau Pinang, and Perak).
 - b. Central region of Peninsular Malaysia: Selangor, Kuala Lumpur, Putrajaya, and Negeri Sembilan.
 - c. East coast region of Peninsular Malaysia: Pahang, Terengganu, and Kelantan.
 - d. Southern region of Peninsular Malaysia: Melaka and Johor.
 - e. East Malaysia: Sarawak, Sabah, and Labuan.

4. Analyze six years of data on registered vehicles in Malaysia, focusing on trends and patterns related to the fuel types of the registered vehicles.
5. Conduct an analysis of six years' worth of data on registered vehicles in Malaysia, highlighting trends and patterns before and after the COVID-19 pandemic. Incorporate the effects of the Movement Control Order (MCO) into the discussion, supported by relevant visuals.

III. Daily Public Transport Ridership Dataset

The daily public transport ridership dataset shows the daily-frequency ridership data for various public transport services across the country, starting from 01 Jan 2019 to 30 Nov 2024. Kindly refer to the URL https://data.gov.my/data-catalogue/ridership_headline to get further information on the dataset. Load the data into R environment. Then, perform data pre-processing and data cleaning and answer the following questions.

6. Using appropriate visuals, show the number of trips aggregated for each month for all the 13 public transport services across the country.
7. Display the 10 days with the highest and lowest ridership for Rapid Bus (KL) in 2023.
8. Using appropriate visuals, analyze the trends and patterns in Rapid Bus (Penang) ridership data for January and February 2022, by aggregating and comparing ridership data for weekdays (Monday to Friday) and weekends (Saturday and Sunday).
9. Using appropriate visuals, display the number of trips aggregated during the festive seasons for the years 2019, 2020, 2021, 2022, and 2023 (LRT Ampang Line, MRT Kajang Line, LRT Kelana Jaya Line, and Monorail Line). The aggregation should include data spanning seven days before and seven days after the first day of each festival.
 - a. Hari Raya Aidilfitri.
 - b. Chinese New Year.
 - c. Deepavali.
 - d. Christmas.
10. Analyze six years of ridership data for the LRT Ampang Line, LRT Kelana Jaya Line, and Monorail Line, focusing on trends and patterns before and after the COVID-19 pandemic. Highlight the impact of the Movement Control Order (MCO) on ridership and support the discussion with relevant visuals.

IV. Submission:

This is a group assignment (a group of four members). The grouping setting will be inherited from Assignment 01.

You are required to submit a zip/rar package which consists of the following items to the eLearn@USM:

- R script (in .R format).
- An assignment report of not more than 15 pages (in pdf format). Only the sample output screen shots and relevant explanation/write-up/description are expected. Also, a cover page which contains your details must be included in your assignment report (not counted as a page limit).

The zip/rar package must be named according to the following notation: CPC351_CPM351_[GroupNumber]_A02. For example, for Group03, they must name the zip/rar package as CPC351_CPM351_Group03_ A02.

One of the group members is required to submit the zip/rar package. Kindly communicate with your group members before the submission to avoid any miscommunication.

The submission deadline is 12 January 2025 (Sunday), 23:59 p.m. Failure to submit the assignment will be a disadvantage to you.

Reference: Kindly state any source of reference in your assignment script should you refer to various sources to complete this assignment.

IMPORTANT: Students who copied or plagiarized other's work or let their work be copied or plagiarized will be given an F grade. The student may be barred from sitting for final exam and reported to the university's disciplinary board.

V. Grading Rubric

This assignment will be graded according to the grading rubric as shown in Table 1. The total will be scaled to 8% of your overall grade.

Table 1: Assignment 02 grading rubric.

	Good (3)	Satisfactory (2)	Poor (1)	Fail (0)
Question 1 (5%)	- Meet all the requirements and contain all the required visuals. The requirements are as follows: - The choice of visual type. - Correctness of information display. - Visual title, colour scheme, visual legend, axis labels, and measurement units. - The R program can be executed, and correct outputs are shown. - Clear and detailed comments are added to the scripts with excellent clarity. - The report includes the screen shots, and explains the results with excellent clarity, comprehensiveness	- Partially meet the requirements. - The R program can be executed, and partially correct outputs are shown. The requirements are as follows: - The choice of visual type. - Correctness of information display. - Visual title, colour scheme, visual legend, axis labels, and measurement units. - Adequate comments are added to scripts with satisfactory clarity. - The report includes the screen shots, and explains the results with satisfactory clarity, comprehensiveness and organization. The description is partially	- Fail to meet the requirements and incorrect outputs are shown. The requirements are as follows: - The choice of visual type. - Correctness of information display. - Visual title, colour scheme, visual legend, axis labels, and measurement units. - The R program cannot be executed, and incorrect outputs are shown - Minimal or no comments is added to the scripts. - The report includes the screen shots, and unclearly or loosely explains the	No submission or late submission.
Question 2 (5%)				
Question 3 (10%)				
Question 4 (15%)				
Question 5 (15%)				
Question 6 (5%)				
Question 7 (5%)				

Question 8 (10%)	and organization. The description is supported by the visuals created or additional visuals • Discussions are well focused, and important points are included.	supported by the visuals created or additional visuals. • Discussions are not comprehensive, and lacking of some important points.	results. The description is not supported by any visuals • Discussion are not well focused, and lacking of important points.	
Question 9 (15%)				
Question 10 (15%)				

~~END OF ASSIGNMENT 02~~